Yohana Bezuayene

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EDUCATION

Minnesota State University, Mankato Mankato, MN

Bachelor of Science in Robotics Engineering. GPA: 3.95/4.00 *Expected May 2027*

Relevant Coursework: Applied Robotics Systems, Mobile Robotics, Automatic Controls, Robotics Programming in C++, Dynamics, Control Systems, Embedded Systems, Microprocessor Engineering I & II

SKILLS

Languages: C++, Python, MATLAB, Assembly, C

Hardware & Tools: Arduino, STM32, Raspberry Pi, LiDAR, Mecanum Wheels, L298N, IR Sensors, Simulink, USART, I2C, Ubuntu, URDF, VS Code

Software: ROS2, Gazebo, MoveIt, OpenCV, Git, Linux, SolidWorks

Concepts: PID Control, SLAM, Path Planning, Inverse Kinematics, Sensor Fusion, Autonomous Navigation

PROJECTS

NAVIS — Navigation and Autonomous Indoor System — Aug 2026 – Present

Senior Design Project | Guide and Delivery Robot

- Designing an autonomous robot that eliminates pre-mapping and pre-configuration requirements, enabling zero-setup deployment in any indoor environment through real-time SLAM-based mapping from uploaded floor plans
- Integrating LiDAR, camera, and mecanum wheels into a ROS2 navigation stack to achieve omnidirectional mobility and dynamic obstacle avoidance in previously unseen environments

Autonomous Navigation System — Q-Bot — Aug 2026 – Present

Applied Robotics Systems | Class Project

- Addressing the challenge of robust indoor autonomy by developing a tiered navigation system combining line-following, LiDAR obstacle detection, computer vision target identification, and payload delivery on a single platform

Autonomous Line-Following Robot — PID Control — Mar 2026 – Apr 2026

- Built a 2WD robot on Arduino Uno with L298N motor driver and 5-channel IR sensor array; implemented a PID controller in C++ with tuned Kp, Ki, Kd parameters to minimize line deviation at high speed
- Designed and wired the full hardware stack, including motor driver, power circuit, battery management, and sensor mount

ROS2 3-DOF Robot Arm Simulation — Gazebo + MoveIt — Feb 2026 – Mar 2026

- Designed a 3-DOF arm with two-fingered gripper in URDF with full collision geometry; simulated physics in Gazebo and configured MoveIt with custom SRDF and KDL kinematics solver, enabling collision-aware motion planning
- Developed a Python ROS2 node for joint state publishing, achieving smooth coordinated motion across all three joints and demonstrating end-to-end integration of simulation and control

EXPERIENCE

Undergraduate Research Assistant Sep 2024 – Nov 2024

Minnesota State University, Mankato *Mankato, MN*

- Contributed to autonomous robotics research using the LEGO RCX 2.0; developed motor control and autonomy demos used to engage high school students in STEM, directly supporting departmental outreach goals

LEADERSHIP & ACTIVITIES

IEEE Robotics & Automation Society (RAS) — MSU Student Branch Chapter Dec 2025 – Present

Founder & President

- Founded the first IEEE RAS student branch chapter (in the state of Minnesota) at MSU, growing membership to an active base and establishing recurring technical workshops and faculty and industry connections

XRP Orbit Odyssey Robotics Competition June 2025

Competitor

- Co-designed and 3D-printed a custom robot chassis and gripper; programmed autonomous navigation and object manipulation in Python; calibrated sensors and motor drivers for reliable competition-ready motion control